Site-to-Site IPsec VPN Configuration Report

Network Infrastructure & Security Lab | Technologies: Cisco Packet Tracer, IPsec, ACL, Crypto Maps

1. Network Topology Overview

This lab implements a secure Site-to-Site IPsec VPN tunnel between two enterprise routers: **DHK-Router** (Dhaka) and **CTG-Router** (Chattogram). The architecture establishes a private, encrypted tunnel across an untrusted public cloud infrastructure, ensuring data confidentiality, integrity, and origin authentication for traffic traveling between local subnets **192.168.1.0/24** and **192.168.20.0/24**.

2. IPsec VPN Implementation Framework

The secure configuration is deployed across 5 core systematic stages:

- **Stage 1: Define Interesting Traffic (ACL)** - Classifies network paths mapping out local-to-remote private communication.
- **Stage 2: ISAKMP Policy (Phase 1)** - Establishes IKE Phase 1 parameter sets including encryption standard (AES-256), pre-shared keys, and Diffie-Hellman parameters.
- **Stage 3: Transform Set (Phase 2)** - Defines data plane encapsulation standards (esp-aes esp-sha-hmac) for operational tunnel traffic.
- **Stage 4: Crypto Map Composition** - Integrates the designated peer address, transform sets, and access lists into a unified crypto instance.
- **Stage 5: Interface Binding** - Attaches the composite crypto map directly to the public-facing physical interfaces.

3. Device Configurations

A. DHK-Router Configuration Commands

```
DHK-Router# configure terminal
DHK-Router(config)# license boot module c1900 technology-package securityk9
DHK-Router# write
DHK-Router# reload

DHK-Router# configure terminal
DHK-Router(config)# access-list 100 permit ip 192.168.1.0 0.0.0.255 192.168.20.0 0.0.0.255
DHK-Router(config)# crypto isakmp policy 20
DHK-Router(config-isakmp)# encryption aes 256
DHK-Router(config-isakmp)# group 5
DHK-Router(config-isakmp)# authentication pre-share
DHK-Router(config-isakmp)# exit

DHK-Router(config)# crypto isakmp key vpnkey address 209.165.200.2
DHK-Router(config)# crypto ipsec transform-set DHK->CTG esp-aes esp-sha-hmac
DHK-Router(config)# crypto map VPNMAP 20 ipsec-isakmp
DHK-Router(config-crypto-map)# set peer 209.165.200.2
DHK-Router(config-crypto-map)# set transform-set DHK->CTG
DHK-Router(config-crypto-map)# set pfs group5
DHK-Router(config-crypto-map)# set security-association lifetime seconds 86400
DHK-Router(config-crypto-map)# match address 100
DHK-Router(config-crypto-map)# interface gigabitethernet 0/1
DHK-Router(config-if)# crypto map VPNMAP
```

B. CTG-Router Configuration Commands

```
CTG-Router# configure terminal
CTG-Router(config)# license boot module c1900 technology-package securityk9
CTG-Router# write
CTG-Router# reload

CTG-Router# configure terminal
CTG-Router(config)# access-list 100 permit ip 192.168.20.0 0.0.0.255 192.168.1.0 0.0.0.255
CTG-Router(config)# crypto isakmp policy 30
CTG-Router(config-isakmp)# encryption aes 256
CTG-Router(config-isakmp)# group 5
CTG-Router(config-isakmp)# authentication pre-share
CTG-Router(config-isakmp)# exit

CTG-Router(config)# crypto isakmp key vpnkey address 209.165.100.2
CTG-Router(config)# crypto ipsec transform-set CTG->DHK esp-aes esp-sha-hmac
CTG-Router(config)# crypto map VPNMAP 30 ipsec-isakmp
CTG-Router(config-crypto-map)# set peer 209.165.100.2
CTG-Router(config-crypto-map)# set transform-set CTG->DHK
CTG-Router(config-crypto-map)# set pfs group5
CTG-Router(config-crypto-map)# set security-association lifetime seconds 86400
CTG-Router(config-crypto-map)# match address 100
CTG-Router(config-crypto-map)# interface gigabitethernet 0/1
CTG-Router(config-if)# crypto map VPNMAP
```

4. Verification and Testing

Connectivity testing was successfully validated via end-to-end ICMP echo requests from host machines across the secure perimeter. Terminal verification tracks confirming structural activation:

```
*Jan 3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
Reply from 192.168.20.10: bytes=32 time<1ms TTL=126
Reply from 192.168.20.10: bytes=32 time<1ms TTL=126
```

The system verification indicates seamless initialization of the encryption environment, confirming that interesting packets reliably activate security policies and establish data tunnels smoothly.